

CSE 4408

System Analysis and Design

Lab 9: Analyzing Systems Using Data Dictionaries

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MedRadar

by team **CookiesAndCaches**

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Project Context

MedRadar is a location-based medicine locator system aimed at helping students and local residents quickly find pharmacies that have the required medicines in stock.

Data Flow Definition

Data Flow Description	
ID	02
Name	Search Query
Description	Contains the customer's search input
Source	Customer (External Entity)
Destination	Process 2 (Handle Search Query)
Type of Data Flow	
☐ File	☐ Screen
☐ Report	☒ Form
☐ Internal	
Associated Data Structure	Search_Query
Volume/Time	~150/day
Comments	Search request for a medicine, with optional category filters. Can be submitted through the system's search interface.

Data Structure Definition

Chosen data structure: Search_Query

Search_Query = Medicine_Name + Location_Info + (Category_Filter) + Customer_ID

Medicine_Name = string

Location_Info = Latitude + Longitude

Category_Filter = { [Antibiotic | Painkiller | Antiseptic | Other]}

Customer_ID = string

Data Element Definition-1

ID	01		
Name	Medicine_Name		
Alias	Med_Name		
Description	The name of the medicine being searched for by the user.		
Element Characteristics			
Length	Variable (max 50)	Data Type	VARCHAR(50)
Format	{ [A-Z a-z 0-9 space '-' ''' '.'] }		
Type	☒ Base ☐ Derived	Default	None
Validation Criteria	• Cannot be NULL. • Must contain at least one alphanumeric character. • Case-insensitive matching during search (not in storage)		
Comments	Supports both generic and brand names; search queries should treat input as case-insensitive.		

Data Element Definition-2

ID	02
Name	Location_Info
Alias	Location
Description	Geographic coordinates of the customer(latitude and longitude)
Element Characteristics	
Length	DECIMAL(9,6)
Data Type	Composite
Format	Latitude: ±DD.dddddd; Longitude: ±DDD.dddddd
Type	☐ Base ☒ Derived
Default	None
Validation Criteria	• Latitude must be between –90.000000 and +90.000000 • Longitude must be between –180.000000 and +180.000000
Comments	Latitude and longitude captured from the customer's device via map API and grouped into a single composite element for search filtering.

Data Element Definition-3

ID	03		
Name	Category_Filter		
Alias	Category_Code		
Description	Optional short code for medicine category in Medicine Master table		
Element Characteristics			
Length	Fixed (3)	Data Type	CHAR(3)
Format	AAA (e.g., “ABT“, “PAI“, “AST“) – where “ABT“ = Antibiotic and similar		
Type	☒ Base	☐ Derived	Default NULL
Validation Criteria	Must exist in the Categories listed in the Medicine Master table.		
Comments	Enables consistent categorization and improves search filtering precision		

Data Store Defintion

ID	D3
Name	Stock_Master
Alias	Pharmacy Stock Table
Description	Stores current stock records for all pharmacies
Data Store Characteristics	
File Type	☒ Computer ☐ Manual
File Format	Relational Table (MySQL)
Associated Data Structure	Medicine_Stock_Record
Primary Key	Stock_ID,
Secondary Keys	Foreign Key : Medicine_ID , Pharmacy_ID
Comments	Updated multiple times daily by pharmacies; indexed for fast search; backed up daily.

Ambiguities Challenges and Discoveries

- Used short 3-letter codes for medicine categories to keep data consistent and save space.
- Stored latitude and longitude with six decimals to ensure precise location data.
- Medicine name searches ignore case, but original spelling is preserved in storage.

Thank you!

Feel free to ask any questions!